Reference Semantics

Wednesday, February 11, 2026 8:57 AM

Goals: Understand how objects behave differently from primitive data. (reference semantics)

Review: · Object type is coded by making a class with instance variables to describe data in an object
· Methods are actions that apply to objects.
· Both are not static, but static functions exist that do not apply on objects.

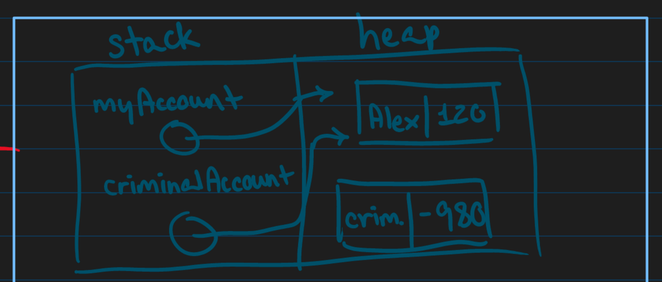
Coding Example (Bank Account)

Inst. Variables: String name, double balance

Methods: constructor called bankAccount
deposit, withdraw, getBalance ← setters set
getters get values of int variables.

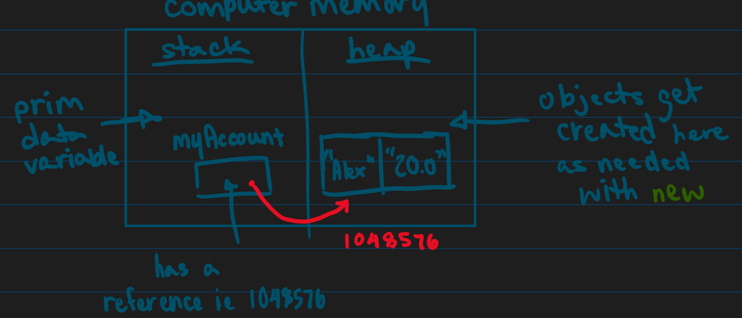
```
public class BankAccount {
    String name;
    double balance;

    public void getName(){
        return name;
    }
    public void setName(String n){
        name = n;
    }
    public double getBalance(){
        return balance;
    }
    public void setBalance(double b){
        balance = b;
    }
    public BankAccount(String n, double b){
        name = n;
        balance = b;
    }
    public void withdraw(double w){
        balance -= w;
    }
    public void deposit(double d){
        balance += d;
    }
    public static void main(String args[]){
        BankAccount myAccount, criminalAccount;
        myAccount = new BankAccount("Alex", 20.0);
        criminalAccount = new BankAccount("BadGuy", 20.0);
        print2Acct(myAccount, criminalAccount);
        myAccount.deposit(100);
        criminalAccount.withdraw(1000.0);
        print2Acct(myAccount, criminalAccount);
        criminalAccount = myAccount;
        print2Account(myAccount, criminalAccount);
    }
    private static void print2Acct(BankAccount a, BankAccount b){
        System.out.println(a.getName() + "has" + a.getBalance());
        System.out.println(b.getName() + "has" + b.getBalance());
```

- static functions can work with objects that we see as parameters.
- methods apply to one object that isn't a parameter.



criminalAccount = myAccount;
- The reference has changed and criminal has myAccount
- The Bad Guy's account has been lost, and has become garbage.

- The Java new operator creates objects



- new BankAccount("Alex", 20); tells Java to set up this object in the heap.
- it will produce a reference to that object.
- the reference says where object likes

variable = thing ; assignment
- figure to right first, put in left.

* We can think of a reference (which is the value in an object variable) as
- an actual machine address
- an arrow pointing to the data (not static)

the variable is "moved" to sit next to the data